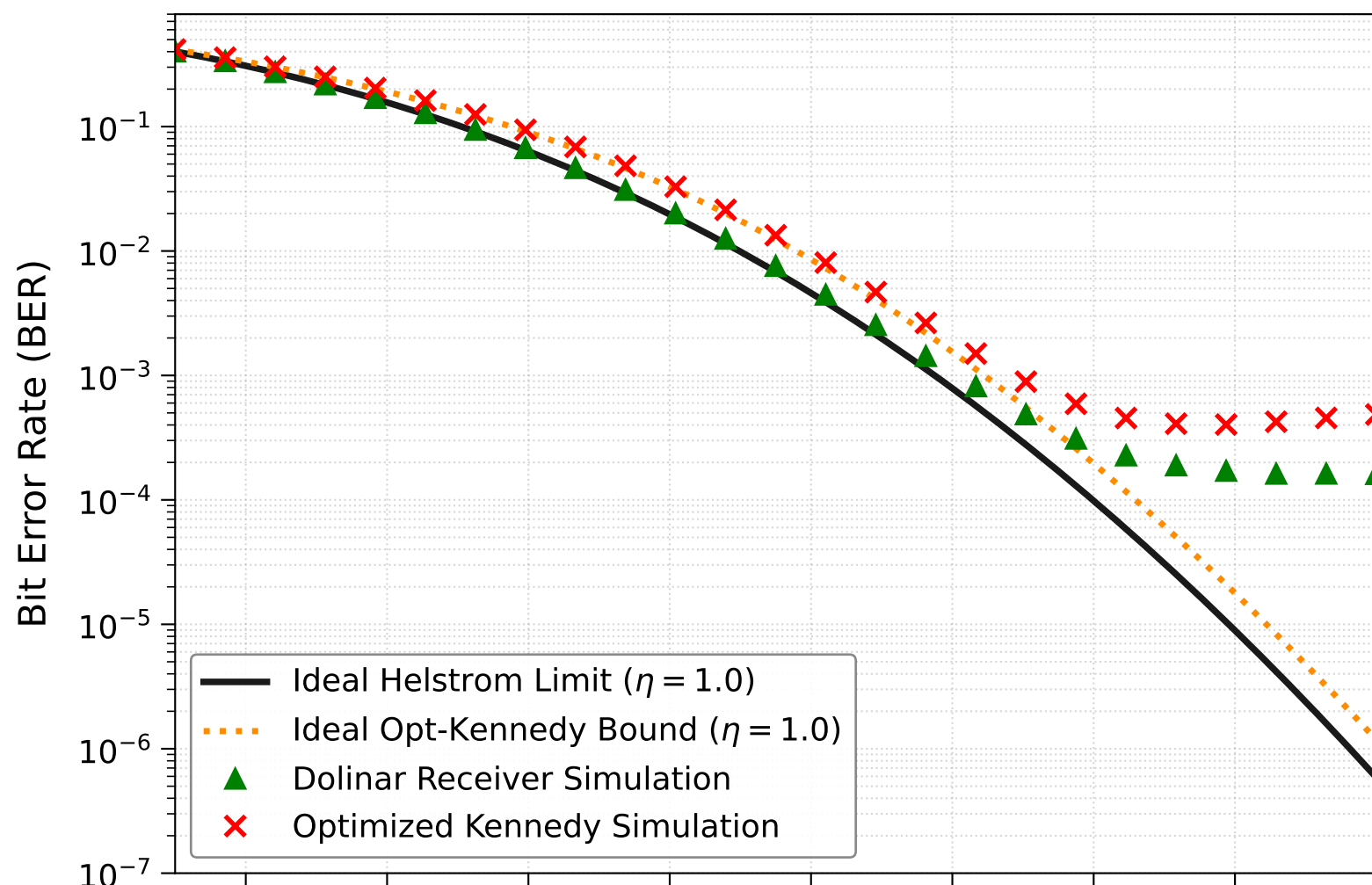


Quantum Receiver Discrimination: Real-World Hardware Comparison (Lab vs. Commercial)

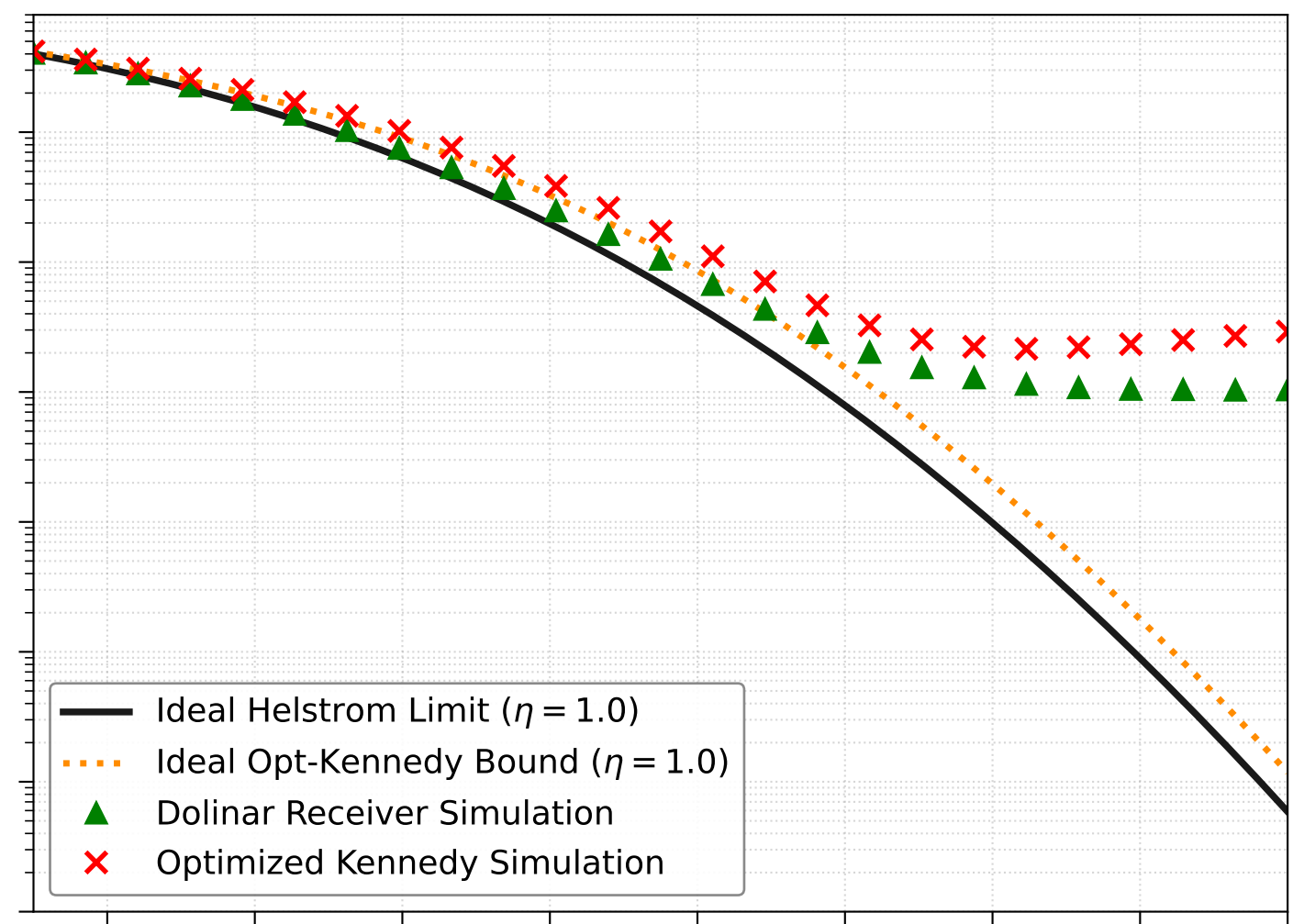
1. Lab State-of-the-Art SNSPD (NIST / JPL)

$(\eta = 0.98, \nu = 10^{-5}, \tau = 0.003T, \delta\phi = 1.0^\circ)$



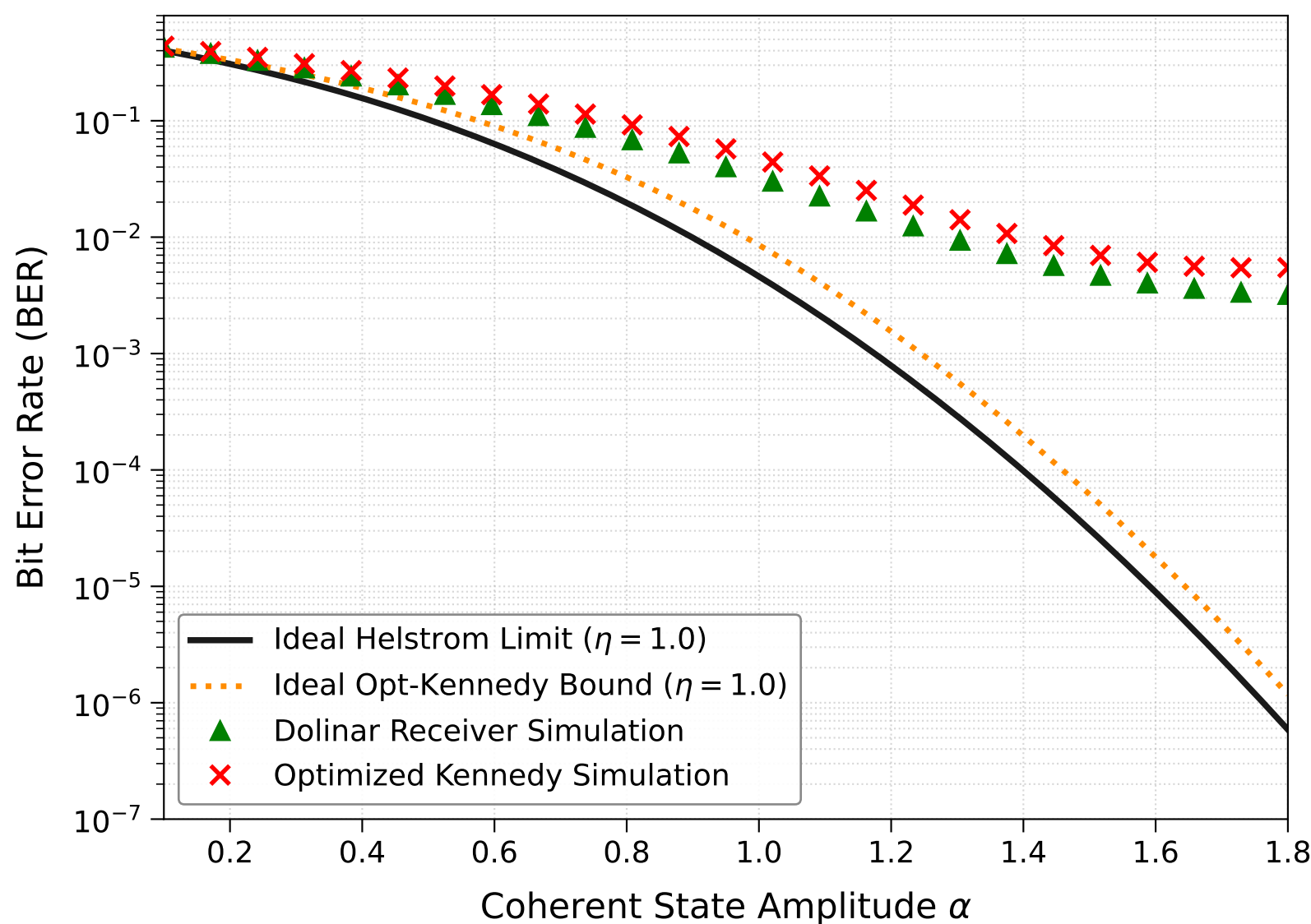
2. Commercial Turnkey SNSPD (Single Quantum)

$(\eta = 0.92, \nu = 2 \times 10^{-4}, \tau = 0.006T, \delta\phi = 2.5^\circ)$



3. Lab APD Benchmark (Becerra et al., NIST 2013)

$(\eta = 0.55, \nu = 1.6 \times 10^{-3}, \tau = 0.014T, \delta\phi = 4.0^\circ)$



4. Commercial InGaAs APD (ID Quantique ID230)

$(\eta = 0.25, \nu = 10^{-2}, \tau = 0.020T, \delta\phi = 8.0^\circ)$

